INCH-POUND

MIL-DTL-55302/38C
20 July 2007
SUPERSEDING
MIL-C-55302/38B (AT)
8 June 1990

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, ELECTRICAL, POLARIZED SHELL, MALE, .100 INCH CONTACT SPACING PIN TERMINAL

Inactive for new design after 4 June 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

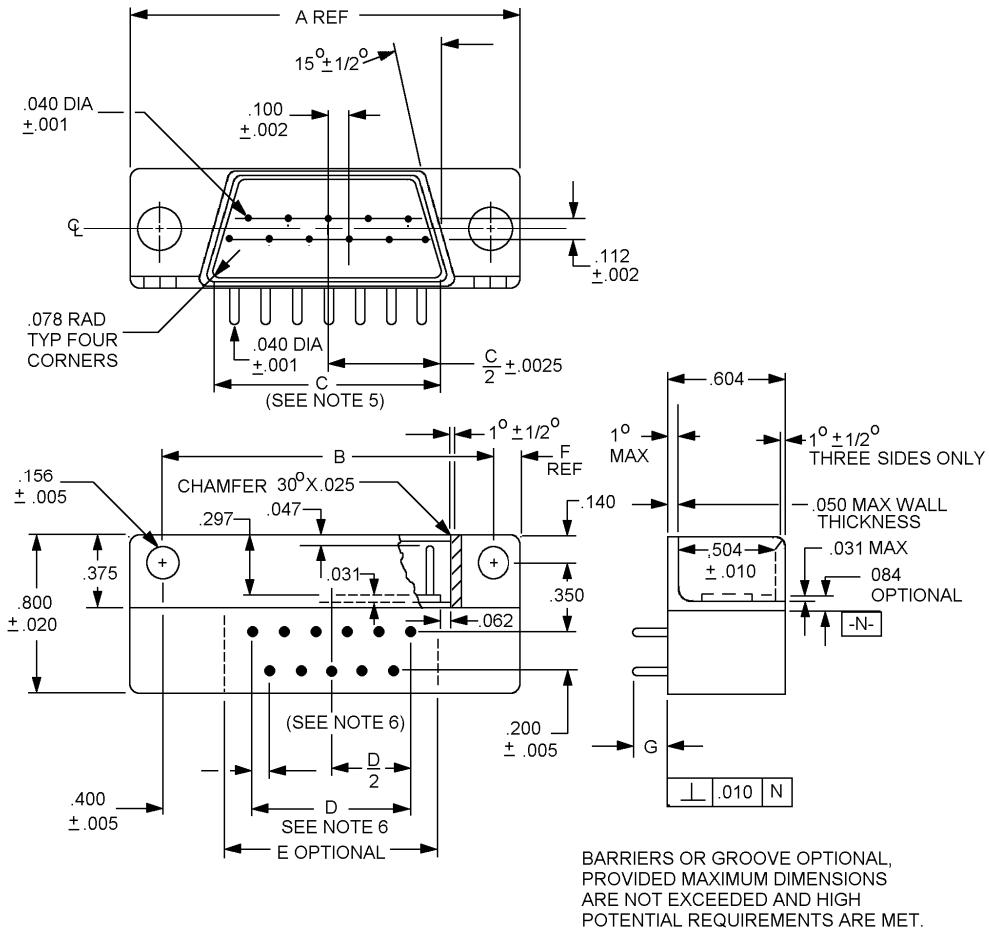


FIGURE 1. Connector, plug, male, .100 inch contact spacing, pin terminal.

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Inches	mm
.001	0.03
.002	0.08
.0025	0.06
.005	0.13
.010	0.25
.020	0.51
.025	0.64
.031	0.79
.040	1.02
.047	1.19
.050	1.27
.062	1.57
.078	1.98
.084	2.13
.100	2.54
.112	2.84
.140	3.56
.156	3.96
.200	5.08
.297	7.54
.350	8.89
.375	9.53
.503	12.80
.604	15.34
.800	20.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.015 (0.38 mm).
4. Break all sharp edges .005 (0.13 mm) minimum.
5 C dimension is measured at the top of the draft angle at the base of the chamfer, at center line of shell.
6. Nominal spacing between any two contacts shall be .100 ± .002 (2.54 ± 0.005 mm); overall tolerance between any span of 23 to 28 contacts through 35 contacts shall be ±.005 (0.13 mm).
7. For dimensions A through G, see table I.
8. Detail contour of insert beyond shell optional, provided maximum dimensions are not exceeded.
9. Number 6 fillister head machine screw recommended for mounting.

FIGURE 1. Connector, plug, male, .100 inch contact spacing, pin terminal – Continued.

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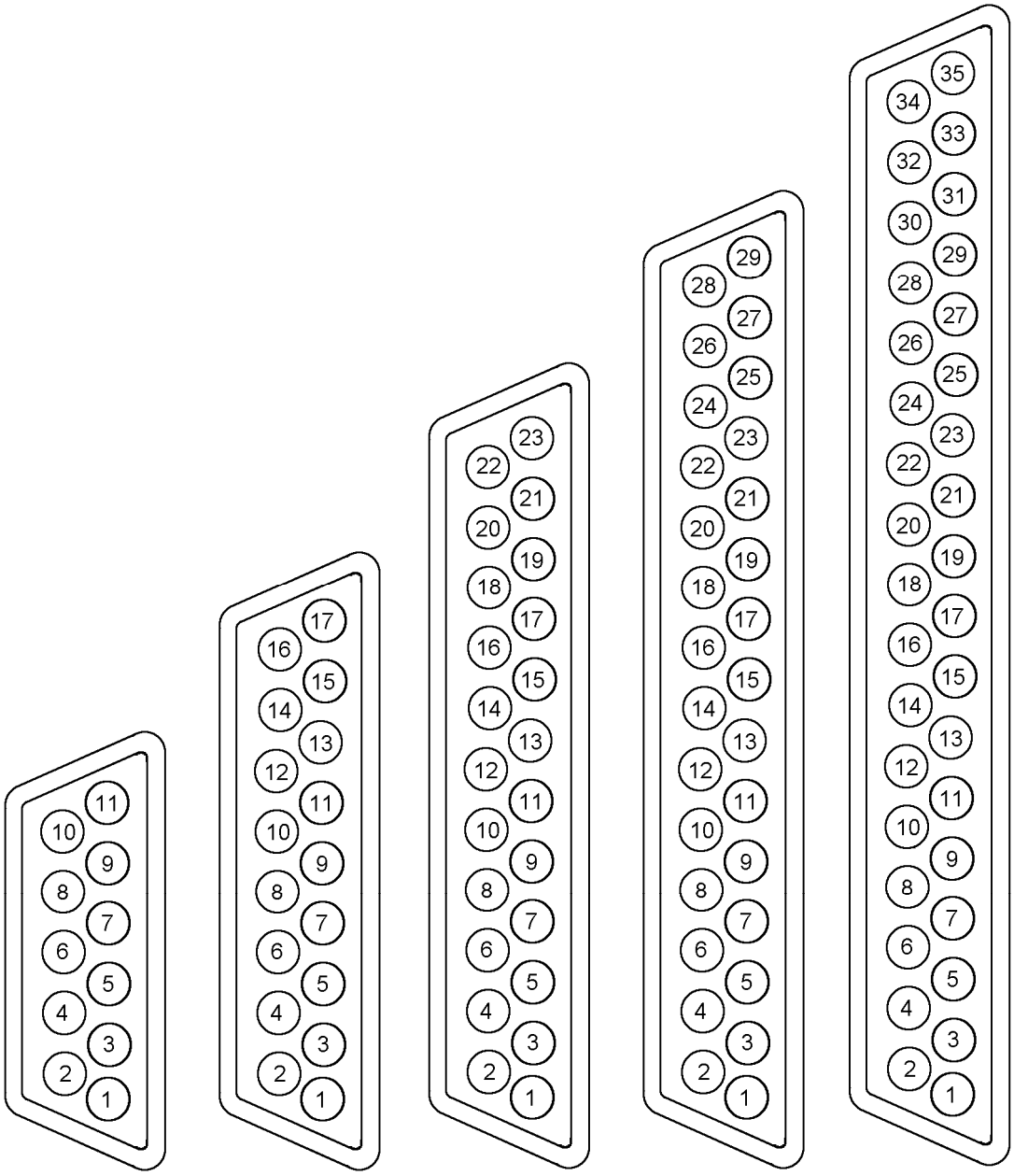


FIGURE 2. Insert arrangement for contact spacing connectors (female engaging face).

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TABLE I. Part or Identifying Number (PIN), dimensions and characteristics.

PIN	Dimensions 1/ 2/							Characteristics		
	A	B ±.010 (0.25)	C ±.005 (0.13)	D	E min	F	G +.015 -.000 (0.38)	Number of contacts	Board thickness	Contact termination
M55302/38-01	2.097 (53.26)	1.800 (45.72)	1.260 (32.00)	1.000 (25.40)	1.250 (31.75)	.149 (3.78)	.125 (3.18)	11	.0625	Pin. right angle
M55302/38-02	2.097 (53.26)	1.800 (45.72)	1.260 (32.00)	1.000 (25.40)	1.250 (31.75)	.149 (3.78)	.125 (3.18)	11	.125	Pin. right angle
M55302/38-03	2.097 (53.26)	1.800 (45.72)	1.260 (32.00)	1.000 (25.40)	1.250 (31.75)	.149 (3.78)	.125 (3.18)	11	.25	Pin. right angle
M55302/38-04	2.721 (69.11)	2.400 (60.96)	1.884 (47.85)	1.600 (40.64)	1.805 (45.85)	.161 (4.09)	.188 (4.78)	17	.0625	Pin. right angle
M55302/38-05	2.721 (69.11)	2.400 (60.96)	1.884 (47.85)	1.600 (40.64)	1.805 (45.85)	.161 (4.09)	.188 (4.78)	17	.125	Pin. right angle
M55302/38-06	2.721 (69.11)	2.400 (60.96)	1.884 (47.85)	1.600 (40.64)	1.805 (45.85)	.161 (4.09)	.188 (4.78)	17	.25	Pin. right angle
M55302/38-07	3.346 (84.99)	3.000 (76.20)	2.509 (63.73)	2.200 (55.88)	2.405 (61.09)	.173 (4.39)	.312 (7.92)	23	.0625	Pin. right angle
M55302/38-08	3.346 (84.99)	3.000 (76.20)	2.509 (63.73)	2.200 (55.88)	2.405 (61.09)	.173 (4.39)	.312 (7.92)	23	.125	Pin. right angle
M55302/38-09	3.346 (84.99)	3.000 (76.20)	2.509 (63.73)	2.200 (55.88)	2.405 (61.09)	.173 (4.39)	.312 (7.92)	23	.25	Pin. right angle
M55302/38-10	3.971 (100.86)	3.600 (91.44)	3.134 (79.60)	2.800 (71.12)	3.005 (76.33)	.186 (4.72)	.125 (3.18)	29	.0625	Pin. right angle
M55302/38-11	3.971 (100.86)	3.600 (91.44)	3.134 (79.60)	2.800 (71.12)	3.005 (76.33)	.186 (4.72)	.188 (4.78)	29	.125	Pin. right angle
M55302/38-12	3.971 (100.86)	3.600 (91.44)	3.134 (79.60)	2.800 (71.12)	3.005 (76.33)	.186 (4.72)	.312 (7.92)	29	.25	Pin. right angle
M55302/38-13	4.596 (116.74)	4.200 (106.68)	3.759 (95.48)	3.400 (86.36)	3.605 (91.57)	.198 (5.03)	.125 (3.18)	35	.0625	Pin. right angle
M55302/38-14	4.596 (116.74)	4.200 (106.68)	3.759 (95.48)	3.400 (86.36)	3.605 (91.57)	.198 (5.03)	.188 (4.78)	35	.125	Pin. right angle
M55302/38-15	4.596 (116.74)	4.200 (106.68)	3.759 (95.48)	3.400 (86.36)	3.605 (91.57)	.198 (5.03)	.312 (7.92)	35	.25	Pin. right angle

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

REQUIREMENTS:

Dimensions and configurations: See figure 1 and table I.

Material:

Trapezoidal shells: Die cast aluminum in accordance with SAE-AMS-QQ-A-591, composition 13, A13, 380, or A380, or ASTM B85, alloy 380.0 or A380.0, cadmium plated, iridescent gold color, type II, class 1 or 2, in accordance with SAE-AMS-QQ-P-416.

Contact identification: Shall be numerical and sequential, in the pattern indicated. See figure 2.

Gold plating: Gold plating shall be in accordance with ASTM B488, type II, class 2.54. Silver underplate shall not be used

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Contact resistance: The average resistance of all contact pairs measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:

Sea level: 1,700 volts rms, 60 Hz, ac.
70,000 feet altitude: 565 volts rms, 60 Hz, ac.

Working voltage: 500 volts rms, maximum.

Contact size: 20.

Current rating, maximum: 7.5 amperes.

Contact current rating: 5 amperes.

Insert arrangement: See figure 2.

Mating connector: MIL-DTL-55302/42.

PIN: See table I.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/42
SAE-AMS-QQ-A-591
SAE-AMS-QQ-P-416
ASTM B85
ASTM B488

CONCLUDING MATERIAL

Custodians:
Army - AT
DLA - CC

Preparing activity:
DLA - CC

(Project: 5935-4693-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .

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